

easier for industries to customise the code according to their needs. IndigoSCADA also makes sure that they work with lower-level languages as much as possible.

Worker's point of view.

Safety of workers and equipment is increased through predefined processes managed by SCADA system. This, thereby, reduces engineering costs, time risk and also provides easy integration with all plant devices. Further, maintenance costs are also reduced through centralised control monitoring that minimises downtime.

Better productivity and quality.

Productivity is increased via analysis of processes used to improve plant and production efficiencies. When it comes to quality, it is improved because of repetitive analysis of process data, which prevents errors even before these occur. Operators are more effective here because it consolidates various plant processes and provides users with a comprehensive overview of operations. Also, mobility solutions in SCADA provide operators with the freedom to observe operations in the first hand no matter where they are.

Extended benefits. IndigoSCADA supports multiple users with different access rights, which makes it easier to control the whole system. IndigoSCADA supports multiple human-machine interface (HMI) windows and multiple dedicated lines on the front-end that make multiprocessing easier. Alarms are centrally-managed, which improves operational effectiveness by not overwhelming operators unnecessarily and also provides real-time event and alarm notifications.

Software architecture

The inter-process communication of IndigoSCADA is based on client/server and publish/subscribe pattern. Inside a process,

HARDWARE REQUIREMENTS

To work with IndigoSCADA, you need a keyboard or mouse, RS232 serial interface and PC or laptop with the following minimum specifications:

Operating system	Minimum requirements		
	Processor	RAM	Graphics
MS Windows Professional XP	800MHz	512GB	XGA 1024×768 16-bit colour depth
MS Windows Server 2003	2.4GHz	1GB	XGA 1024×768 16-bit colour depth
MS Windows Vista	1GHz	1GB	XGA 1024×768 16-bit colour depth
MS Windows 7	1GHz	1GB	XGA 1024×768 16-bit colour depth

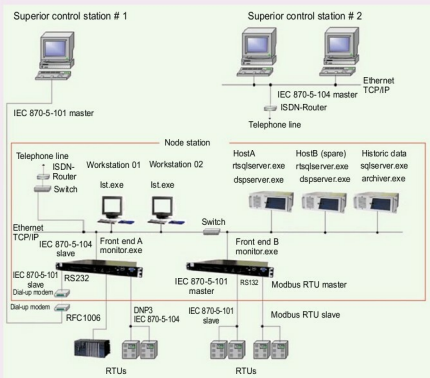


Fig. 2: Architecture of IndigoSCADA with communication protocols

the observer pattern (a subset of publish/subscribe pattern) is used. It uses signals and slots language construct, introduced in Qt, which makes it easy to implement the observer pattern. HMI designer of IndigoSCADA acts as a tool that allows you to use data for generating management reports. In this, you can insert and edit elements such as widgets, buttons and containers. All of these are highly-customisable and can be manipulated with ease.

Being a volunteer-driven project, IndigoSCADA welcomes contributions in the form of code, bug fixes,

bug reports, contributed documentation, advocacy and supporting other users on IndigoSCADA forum.

Also, there are a lot of financial contributors providing funding and sponsorships to IndigoSCADA. This ensures that when you work on Indigo SCADA, there are many helping hands to resolve your issue in many ways, thereby making IndigoSCADA a conducive SCADA software. **EFY**

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